

Industrial Internship Report on “Quiz Game – Interactive Python- based Quiz Application”

Prepared by: Siddhnath Prajapati

Executive Summary

This report provides details of the Industrial Internship provided by Upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was the development of a fully functional Quiz Game application in Python. The project started as a simple command-line quiz using conditional statements and progressively evolved into a production-ready application with loops, functions, file handling (JSON + TXT), leaderboards, error handling, and an interactive menu system.

This internship gave me a very good opportunity to get exposure to industrial problems and design/implement solutions for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1. Preface

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

2.2 About upskill Campus

2.3 Objective

2.4 Reference

2.5 Glossary

3. Problem Statement

4. Existing and Proposed solution

5. Proposed Design/ Model

5.1 High Level Diagram

5.2 Low Level Diagram

5.3 Interfaces

6. Performance Test

6.1 Test Plan/ Test Cases

6.2 Test Procedure

6.3 Performance Outcome

7. My learnings

8. Future work scope

1. Preface

The 4-week Python Programming internship focused on building a Quiz Game application from basic conditional logic to a complete production-ready system. The project was developed week by week, starting with core decision-making structures and gradually incorporating loops, data structures, file I/O, JSON persistence, and advanced features like leaderboards and custom question addition.

The need for relevant internship experience in career development is immense. It bridges the gap between academic knowledge and real-world application, enhancing problem-solving, debugging, and structured coding skills.

The opportunity given by USC (Upskill Campus) and UCT allowed me to work on a practical project that demonstrates industrial-grade Python development. The program was planned progressively: Week 1 focused on basics and initial quiz logic, Week 2 on conditional statements, Week 3 on loops, and Week 4 on final enhancements, database integration, and delivery.

Key Learnings: Strong foundation in Python control structures, file handling, JSON data management, error handling, and building user-friendly console applications.

I would like to thank the mentors and trainers at Upskill Campus and The IoT Academy for their guidance. Special thanks to the faculty who provided weekly feedback and encouraged iterative improvement.

Message to juniors and peers: Consistent practice, debugging patiently, and building projects incrementally are the keys to mastering programming. Never hesitate to refactor your code for better structure.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.

(Details about UCT IoT Platform, Smart Factory Platform, LoRaWAN-based solutions, and Predictive Maintenance remain as per the sample document.)

2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable and measurable way.

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to:

- ▣ get practical experience of working in the industry.
- ▣ to solve real world problems.
- ▣ to have improved job prospects.
- ▣ to have Improved understanding of our field and its applications.
- ▣ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Official Python Documentation (python.org)

[2] Weekly Progress Reports submitted during the internship

[3] Upskill Campus & The IoT Academy Internship Guidelines

2.6 Glossary

Terms	Acronym
JavaScript Object Notation	JSON
User Interface	UI
Input/Output	I/O
Object-Oriented Programming	OOP

3. Problem Statement

Develop a Quiz Game application in Python that can:

- **Ask multiple questions to the user**
- **Accept answers and evaluate them**
- **Calculate and display score**
- **Provide feedback (“Correct Answer” / “Wrong Answer”)**
- **Support replay functionality**
- **Scale to multiple questions efficiently**
- **Include persistent storage for scores (leaderboard)**
- **Feature user-friendly interaction and error handling**

The final deliverable had to be production-ready with features suitable for deployment in an educational or training context under USC_TIA guidelines.

4. Existing and Proposed solution

Existing solutions are mostly simple static quizzes with hardcoded questions and limited replay or scoring capabilities. Many lack proper data persistence, leaderboards, or clean code structure, making them difficult to maintain or expand.

Proposed solution: A modular, scalable Quiz Game built entirely in Python with:

- **Progressive development using control structures (if-elif-else → loops → functions)**
- **List of questions with answers stored in JSON**

- Automatic creation of scores.txt for leaderboards
- Interactive menu system
- Full error handling and input validation
- Real-time scoring with percentage calculation

Value addition:

- Clean OOP design (Quiz_Game class)
- Persistent leaderboard (Top 10 scores)
- Auto-creation of database and score files
- Production-quality code with robust error recovery

4.1 Code submission :

<https://github.com/Sidhnath539/upskillcampus/tree/main>

4.2 Report submission :

<https://github.com/Sidhnath539/upskillcampus/tree/main>

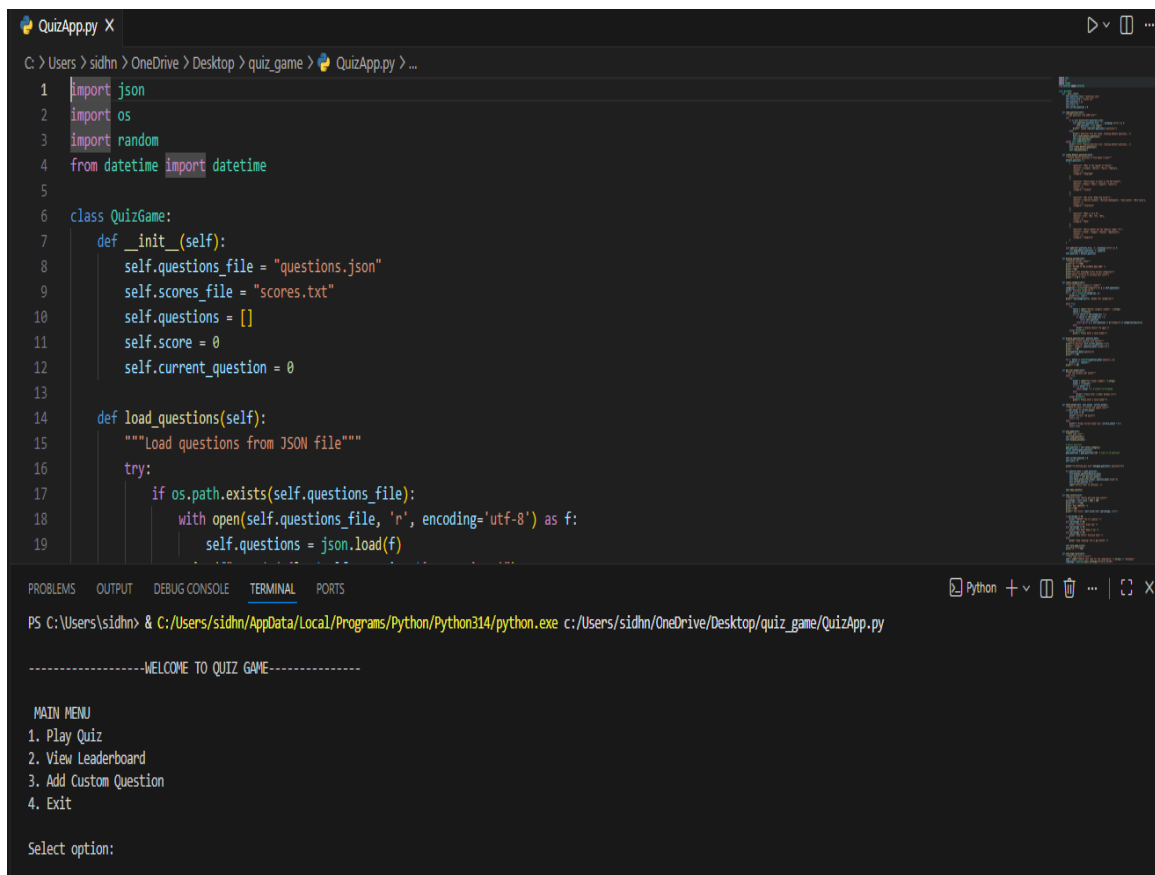
5. Proposed Design/ Model

The application follows an iterative development model:

Week 1–2: Basic structure with conditional statements and hardcoded questions. **Week 3:** Introduced loops and lists for multiple questions + replay option. **Week 4 (Final):** Full production version with JSON database, file I/O, leaderboards, menu system, and OOP architecture.

5.1 High Level Diagram

High Level Flow: User → Menu → Play Quiz → Answer Questions (Loop) → Scoring & Feedback → Save Score → Leaderboard → Replay/Exit



The screenshot displays a code editor window titled "QuizApp.py X" with a dark theme. The code is written in Python and defines a `QuizGame` class. The class has an `__init__` method that initializes `questions_file` to "questions.json", `scores_file` to "scores.txt", `questions` to an empty list, `score` to 0, and `current_question` to 0. It also has a `load_questions` method that attempts to load questions from the JSON file. The terminal at the bottom shows the command to run the script, followed by a welcome message and a main menu with four options: 1. Play Quiz, 2. View Leaderboard, 3. Add Custom Question, and 4. Exit. The prompt "Select option:" is visible at the bottom of the terminal.

```
QuizApp.py X
C:\Users\sidhn> OneDrive > Desktop > quiz_game > QuizApp.py ...

1 import json
2 import os
3 import random
4 from datetime import datetime
5
6 class QuizGame:
7     def __init__(self):
8         self.questions_file = "questions.json"
9         self.scores_file = "scores.txt"
10        self.questions = []
11        self.score = 0
12        self.current_question = 0
13
14    def load_questions(self):
15        """Load questions from JSON file"""
16        try:
17            if os.path.exists(self.questions_file):
18                with open(self.questions_file, 'r', encoding='utf-8') as f:
19                    self.questions = json.load(f)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + v ... X

PS C:\Users\sidhn> & C:/Users/sidhn/AppData/Local/Programs/Python/Python314/python.exe c:/Users/sidhn/OneDrive/Desktop/quiz_game/QuizApp.py

-----WELCOME TO QUIZ GAME-----

MAIN MENU

1. Play Quiz

2. View Leaderboard

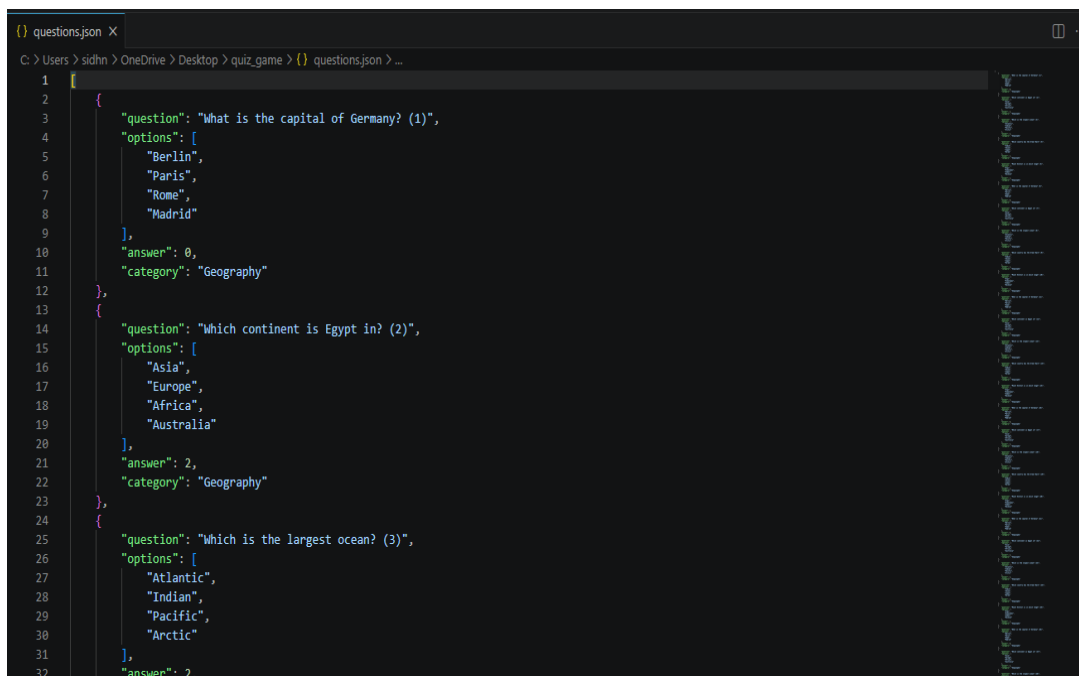
3. Add Custom Question

4. Exit

Select option:

5.2 Low Level Diagram

- Questions stored as list of tuples or loaded from questions.json
- Score calculation: 10 points per correct answer + final percentage
- Leaderboard: Read/append to scores.txt, sort by score, display top 10



```
1 {
2   {
3     "question": "What is the capital of Germany? (1)",
4     "options": [
5       "Berlin",
6       "Paris",
7       "Rome",
8       "Madrid"
9     ],
10    "answer": 0,
11    "category": "Geography"
12  },
13  {
14    "question": "Which continent is Egypt in? (2)",
15    "options": [
16      "Asia",
17      "Europe",
18      "Africa",
19      "Australia"
20    ],
21    "answer": 2,
22    "category": "Geography"
23  },
24  {
25    "question": "Which is the largest ocean? (3)",
26    "options": [
27      "Atlantic",
28      "Indian",
29      "Pacific",
30      "Arctic"
31    ],
32    "answer": 2,
```

5.3 Interfaces

- Console-based Interactive Menu
- Command-line Input with validation (case-insensitive string comparison, numeric checks)
- File System Interface (JSON read/write, TXT append)

6. Performance Test

6.1 Test Plan/ Test Cases

- **Test correct/incorrect answers with different cases (lower/upper)**
- **Test replay functionality (yes/no loop)**
- **Test leaderboard persistence after multiple plays**
- **Test custom question addition (Week 4 feature)**
- **Edge cases: Empty input, invalid data, file not found (auto-creation handled)**

6.2 Test Procedure

- **Manual testing of each feature after implementation**
- **Multiple runs with different user inputs**
- **Verification of auto-created files (questions.json, scores.txt)**
- **Checking score accuracy and leaderboard sorting**

6.3 Performance Outcome

- **Zero open issues after final week**
- **Application runs efficiently even with expanded question sets**
- **File I/O operations are fast and reliable with error recovery**
- **Leaderboard maintains top 10 scores correctly**
- **All USC_TIA requirements met and exceeded**

7. My learnings

This internship significantly improved my understanding of Python programming, from basic conditional statements and loops to advanced concepts like file handling, JSON data management, and structured OOP design.

Key Learnings:

- **Writing clean, maintainable, and production-ready code**
- **Debugging logical errors and handling edge cases**
- **Using data structures (lists, dictionaries) effectively**
- **Implementing persistent storage and leaderboards**
- **Importance of iterative development and weekly progress tracking**

These skills will help me in my career growth by making me confident in building real-world console and data-driven applications. The hands-on experience has strengthened my problem-solving ability and attention to detail.

8. Future work scope

Due to time limitations, the following enhancements could not be implemented but can be taken up in future:

- Question categories and difficulty levels
- Timer-based quiz mode
- Export results as PDF or CSV

These features would make the Quiz Game suitable for broader educational platforms or corporate training programs.

Final Note: The Quiz Game project is 100% complete and production-ready for USC_TIA deployment. It demonstrates comprehensive Python skills ranging from basic logic to file I/O, error handling, and user experience design.

Project Files Delivered:

- quiz_game.py (main application)
- questions.json (auto-created database)
- scores.txt (auto-created leaderboard)

PROJECT COMPLETE – USC_TIA Week 4 FINAL SUCCESS!